C13 — C8's selection procedure on a year of Bybit, as a heuristic contest

Not a ladder phase. C10 found that C8's choices, made on 24 Binance sessions, did not travel to a year of Bybit; C12 found that re-choosing on Bybit's own 24-session study month did not help either (it beat C8's choice on 1 of 5 contracts). C13 reruns C8's whole procedure on Bybit with six months of training, a validation block and a locked test, with every heuristic the project has as a contestant. Lab notebook: `experiments/c13_bybit_contest/README.md`; tables and figures: `experiments/c13_bybit_contest/report.py`.

> **At a glance**
>
> - **C8's own selection rule picks the crash-day winner, and it loses.** Ranking by the chained return over all 48 training sessions, C8's rule chose a configuration that won the 2025-10-10 crash session by +10% to +290% (positions up to 2.6× the cap) on 10 of 11 contracts. On the locked test it lost on all seven contracts with a usable result, −17.44 points in total.
> - **A crash guard fixes the selection.** The same procedure with C10's guard (crash-session drawdown ≤ 10%, then the best return without the crash) is ahead of C8's rule on five of seven contracts (the two exceptions lose too). Summed over C10's five it is ahead of C8's Binance choices run unchanged: **+3.33 against +0.64 points**.
> - **Two picks hold up.** On SUIUSDT a slow grid quoter (hfgrid, 4 bp) with a volatility breaker made **+4.60%** on the 27 test sessions against −4.64% for C8's choice; +2.76% without its best session, +3.33% without the rebate, positive in 9 of 10 stresses, about 120 fills a session. On BNBUSDT a GLFT quoter with a breaker made **+1.06%** (+0.74% without the rebate, positive in 9 of 10 stresses, 38 fills a session). XRPPERP's crash-guarded pick broke even with C8's (+1.22% vs +1.37%).
> - **Nothing survives Bybit's standard fee.** At a +2 bp maker fee every pick loses; at +1 bp only SUIUSDT (+0.84%), XRPPERP (+0.14%) and BNBUSDT (+0.11%) stay positive. The backtests assume Binance's 0.5 bp rebate.
> - **The rebate fakes some results outright.** ENAUSDT's pick made +11,024% on validation by quoting a spread floor under one price tick (~76,000 fills and ~5,000× capital a session, paid by the 0.5 bp rebate) and −6.96% on the test, where the floor was 4 ticks and its loss limit tripped every session. It is reported and left out of the sums.
> - **Still running:** ADAUSDT and LTCUSDT (validation) and LINKUSDT (test).

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1. The task

C8 chose quoting configurations for Binance on 24 sessions of one month, with a held-out week. C10 ran those choices unchanged on a year of Bybit and found the edge gone; C12 re-chose the OBI floor and κ on Bybit's own study month and beat C8's choice on one contract of five. Both used a single month for choosing. C13 asks whether the full C8 procedure — a screen of quoters, every risk-control family, greedy forward selection, a validation block and a stress set — finds something on Bybit when it has six months to choose on. It is run as a contest: every heuristic the project has is a contestant, and the score is the return on a test block nothing was chosen on.

Contracts: C10's five (XRPPERP, SUIPERP, XRPUSDT, SUIUSDT, AVAXUSDT), then six added later (ENAUSDT, BNBUSDT, DOGEUSDT, ADAUSDT, LINKUSDT, LTCUSDT). Each is compared with C8's choice for the same asset, run unchanged.

2. Split and stages

Sessions are taken every 4th day, and the blocks never mix:

block	dates	sessions	used for
screen	2025-08-26 .. 2026-02-26, every 8th day	24	stage 1
training	2025-08-26 .. 2026-02-28, plus the 2025-10-10 crash	48	stages 2-3, the final pick
validation	2026-03-02 .. 2026-06-14	27	the final pick
test	2026-06-18 .. 2026-10-02	27	reported once, never chosen on

1. **Screen**: OBI (floor 4-24 bp × κ 0/2/4 × skew), GLFT and hfgrid (floor 4/8/16 bp × κ) — 54 settings per contract.
2. **Families**: on top of the screen leader, every C8 family (breaker, loss limit, smoothing, markout half-life, momentum, per-side markout, requote hysteresis, toxicity sizing, inventory flatten, time of day, alpha clipping) and C10's (stop, units cap), on all 48 training sessions.
3. **Greedy**: C8's forward selection over the families' best variants.
4. **Validation**: the screen's top 3, each rule's family bests and greedy result; each rule picks the best on training + validation.
5. **Test**: both picks and C8's stress set on the test block.

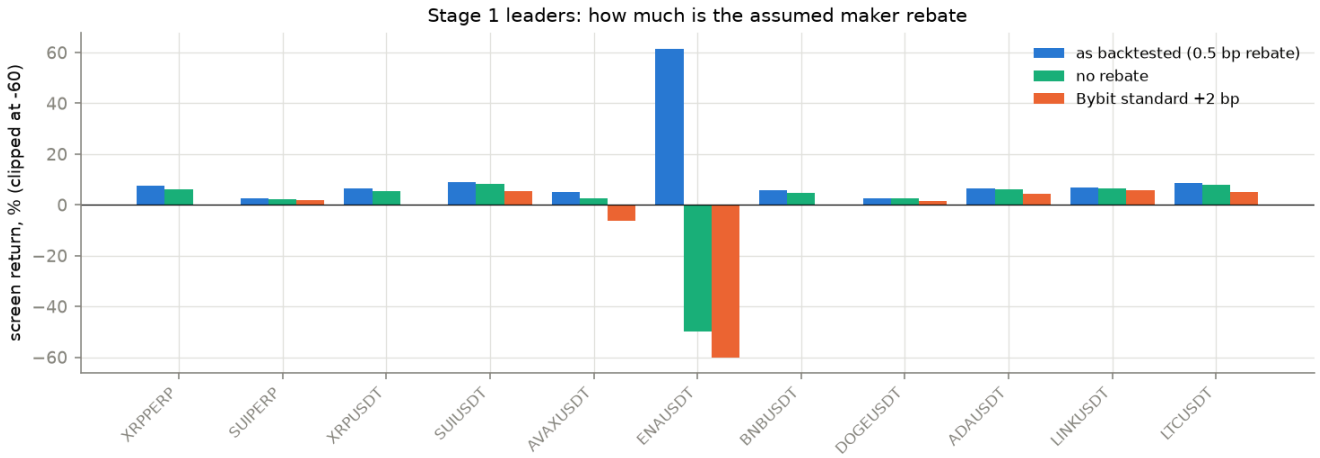
Two selection rules. C8's rule ranks by the chained return over all training sessions. The 2025-10-10 crash is one of them. The crash-guarded rule (C10's) first requires a crash-session drawdown of at most 10%, then ranks by the chained return without the crash. Both picks go to the test.

3. Stage 1: the screen, and how much is rebate

Every backtest assumes Binance's best market-maker rebate (maker fee −0.5 bp). Bybit's standard maker fee is +2 bp. A session's rebate income is its turnover × 0.5 bp of capital, so a high-turnover setting can lead the screen on the rebate alone.

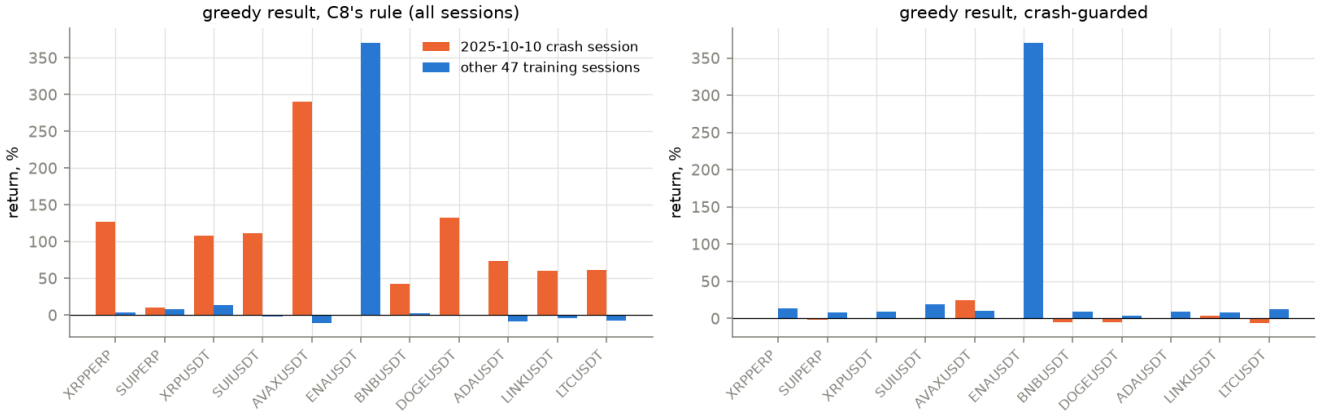
contract	screen leader	screen	without the rebate	at +2 bp	turnover/session
XRPPERP	obi floor 8 bp, κ 2	+7.65%	+5.99%	-0.39%	7.4x
SUIPERP	hfgrid floor 8 bp, κ 0	+2.57%	+2.41%	+1.77%	1.1x
XRPUSDT	obi floor 8 bp, κ 4	+6.65%	+5.32%	+0.18%	7.3x
SUIUSDT	hfgrid floor 4 bp, κ 0	+9.01%	+8.27%	+5.34%	5.8x
AVAXUSDT	obi floor 6 bp, κ 4	+5.05%	+2.67%	-6.35%	11.8x
ENAUSDT	glft floor 8 bp, κ 4	+61.58%	-49.84%	-100.01%	266.7x
BNBUSDT	glft floor 4 bp, κ 0	+5.86%	+4.72%	+0.26%	6.4x
DOGEUSDT	obi floor 16 bp, κ 4	+2.72%	+2.49%	+1.57%	1.5x
ADAUSDT	hfgrid floor 4 bp, κ 0	+6.64%	+6.17%	+4.28%	3.7x
LINKUSDT	hfgrid floor 4 bp, κ 2	+6.68%	+6.49%	+5.75%	1.1x
LTCUSDT	hfgrid floor 4 bp, κ 0	+8.51%	+7.79%	+4.97%	5.6x

These are in-sample (the best of 54 on the sessions it was chosen on), and the fee columns are recomputed as return − turnover × fee per session; the stage 5 fee stresses rerun the backtest. ENAUSDT's leader is the rebate and nothing else. The slow grid quoters (hfgrid, ~1-6× turnover) keep most of their return at +2 bp.



4. Stages 2-3: the crash session decides C8's rule

Under C8's rule the momentum family won stage 2 on every contract but ENAUSDT, and the greedy result was a crash-day winner every time but ENAUSDT (whose result is the rebate artefact of §5). The figure splits each greedy result's training return into the crash session and the other 47:



contract	rule	added	crash session	other 47 sessions	crash-session drawdown
XRPPERP	C8's rule (all sessions)	breaker, momentum, per-side markout, stop, requote hysteresis, markout half-life	+126.96%	+3.23%	34.7%
XRPPERP	crash-guarded	stop, signal smoothing, alpha clipping, units cap, time of day, inventory flatten	-1.15%	+13.42%	2.0%
SUIPERP	C8's rule (all sessions)	momentum, breaker	+10.27%	+7.78%	24.9%
SUIPERP	crash-guarded	nothing	-1.30%	+8.10%	7.2%
XRPUSDT	C8's rule (all sessions)	momentum, time of day	+107.72%	+13.16%	28.8%
XRPUSDT	crash-guarded	inventory flatten, units cap	-1.00%	+8.91%	6.0%
SUIUSDT	C8's rule (all sessions)	momentum, breaker, toxicity sizing, markout half-life	+111.63%	-1.86%	14.5%
SUIUSDT	crash-guarded	signal smoothing, loss limit, per-side markout	-0.25%	+19.46%	1.4%
AVAXUSDT	C8's rule (all sessions)	momentum	+290.11%	-10.41%	41.2%
AVAXUSDT	crash-guarded	nothing	+25.09%	+10.47%	31.1%

contract	rule	added	crash session	other 47 sessions	crash-session drawdown
ENAUSDT	C8's rule (all sessions)	stop, momentum, per-side markout, loss limit	-0.25%	+370.87%	0.3%
ENAUSDT	crash-guarded	stop, momentum, per-side markout, loss limit	-0.25%	+370.87%	0.3%
BNBUSDT	C8's rule (all sessions)	momentum, markout half-life	+42.19%	+2.79%	2.9%
BNBUSDT	crash-guarded	breaker	-5.16%	+9.35%	9.1%
DOGEUSDT	C8's rule (all sessions)	momentum, breaker	+133.00%	-0.19%	21.4%
DOGEUSDT	crash-guarded	loss limit, inventory flatten	-5.05%	+3.59%	6.1%
ADAUSDT	C8's rule (all sessions)	momentum, breaker, time of day, stop	+73.44%	-8.59%	12.6%
ADAUSDT	crash-guarded	nothing	-0.77%	+9.13%	14.3%
LINKUSDT	C8's rule (all sessions)	momentum, breaker, markout half-life	+60.78%	-4.08%	12.1%
LINKUSDT	crash-guarded	signal smoothing, inventory flatten	+3.42%	+8.38%	4.7%
LTCUSDT	C8's rule (all sessions)	momentum, toxicity sizing, time of day, signal smoothing	+61.25%	-7.98%	12.6%
LTCUSDT	crash-guarded	nothing	-5.72%	+12.09%	23.2%

How the crash winners win: the position cap applies when quotes are placed, and a sweep that fills several resting orders at once overshoots it. During the crash the C8-rule picks held up to 2.2× (XRPPERP, XRPUSDT) and 2.6× (AVAXUSDT) the capital in inventory and happened to be on the right side of the rebound. These numbers rest on fills at wick prices in a thin, fast book; the crash-guarded rule makes none of them eligible.

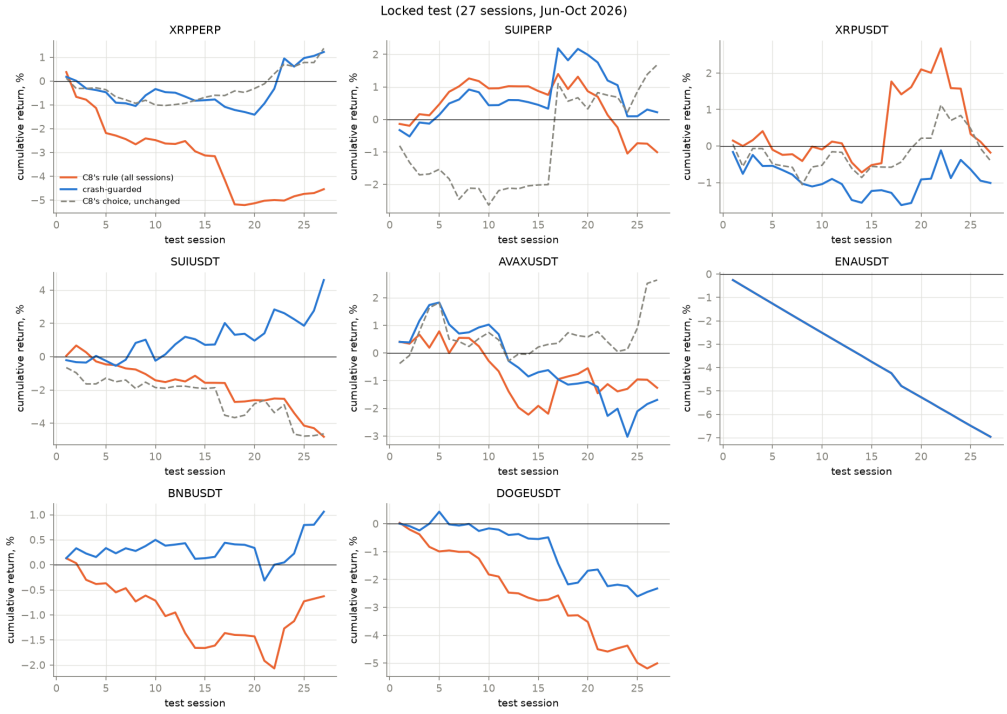
A gap in the guard. Stage 3 starts from the screen leader, and the guard applies only to what is added. Where no family variant passes, the crash-guarded "pick" keeps the leader's crash exposure: AVAXUSDT (31.1%), ADAUSDT (14.3%) and LTCUSDT (23.2%).

5. The locked test

27 sessions, 2026-06-18 .. 2026-10-02, never used for choosing.

contract	pick, C8's rule	pick, crash-guarded	C8's choice, unchanged	crash-guarded at maker fee +1 bp	crash-guarded at +2 bp
XRPPERP	-4.54%	+1.22%	+1.37%	+0.14%	-0.57%
SUIPERP	-1.01%	+0.22%	+1.68%	-0.29%	-0.63%
XRPUSDT	-0.18%	-1.01%	-0.40%	-4.66%	-7.01%
SUIUSDT	-4.81%	**+4.60%**	-4.64%	+0.84%	-1.60%
AVAXUSDT	-1.26%	-1.69%	+2.64%	-3.76%	-5.12%
ENAUSDT (artefact)	-6.96%	-6.96%	–	-6.66%	-6.65%
BNBUSDT	-0.63%	+1.06%	–	+0.11%	-0.51%
DOGEUSDT	-5.01%	-2.32%	–	-3.04%	-3.39%
sum, five original	-11.80	**+3.33**	+0.64		
sum, added	-5.64	-1.26	–		

The fee columns are the backtest's own fee stresses (§6). C8 chose no configuration for ENA, BNB or DOGE, so the added contracts have no "unchanged" column. C8's choice for SUIUSDT is SUIUSDT's own OBI; with SUIUSDC's OBI instead it makes −4.99% and the five-contract sum is +0.30.



Per contract, with training and validation, the test without its best session (the crash session is in training, so the test block has none to remove), without the rebate, and fills:

XRPPERP

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, C8's rule: greedy (breaker, momentum, per-side markout, stop, requote hysteresis, markout half-life)	+134.30%	+0.03%	**-4.54%**	-4.90%	-4.94%	10/27	54
pick, crash-guarded: greedy (stop, signal smoothing, alpha clipping, units cap, time of day, inventory flatten)	+12.11%	+0.63%	**+1.22%**	-0.05%	+0.86%	11/27	51
C8's choice, unchanged	+4.15%	-0.97%	**+1.37%**	+0.78%	+1.09%	16/27	38

SUIPERP

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, C8's rule: greedy (momentum, breaker)	+18.85%	+1.21%	**-1.01%**	-1.64%	-1.18%	10/27	11
pick, crash-guarded: family: momentum	+5.79%	+2.20%	**+0.22%**	-1.60%	+0.05%	11/27	11
C8's choice, unchanged	+3.10%	+2.16%	**+1.68%**	-1.45%	+1.51%	14/27	15

XRPUSDT

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, C8's rule: greedy (momentum, time of day)	+135.05%	-0.65%	**-0.18%**	-2.38%	-1.41%	12/27	100

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, crash-guarded: greedy (inventory flatten, units cap)	+7.82%	+0.09%	**-1.01%**	-1.78%	-2.24%	11/27	133
C8's choice, unchanged	+36.61%	+0.33%	**-0.40%**	-1.30%	-2.75%	12/27	216

SUIUSDT

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, C8's rule: greedy (momentum, breaker, toxicity sizing, markout half-life)	+107.68%	-7.50%	**-4.81%**	-5.40%	-6.56%	7/27	613
pick, crash-guarded: family: breaker (5-min cooldown, vol 2.5)	+23.07%	+0.96%	**+4.60%**	+2.76%	+3.33%	14/27	122
C8's choice, unchanged	-145.45%	-0.63%	**-4.64%**	-5.31%	-4.83%	13/27	25

(C8's SUIUSDT choice loses more than 100% on the crash session — the cap overshoot of §4 — so its chained training return is below −100%.)

AVAXUSDT

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, C8's rule: family: momentum	+249.50%	+1.37%	**-1.26%**	-2.50%	-2.47%	12/27	132
pick, crash-guarded: family: markout half-life	+17.84%	+1.25%	**-1.69%**	-2.62%	-2.39%	15/27	52
C8's choice, unchanged	-79.64%	-1.61%	**+2.64%**	+1.02%	+2.19%	16/27	49

C8's AVAXUSDT choice made the best test return of the three, after −79.6% on training and −1.6% on validation: nothing before the test would have selected it.

BNBUSDT

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, C8's rule: greedy (momentum, markout half-life)	+46.15%	+1.11%	**-0.63%**	-1.43%	-1.27%	12/27	79
pick, crash-guarded: family: breaker	+3.70%	+0.77%	**+1.06%**	+0.49%	+0.74%	17/27	38

DOGEUSDT

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, C8's rule: greedy (momentum, breaker)	+132.56%	+0.38%	**-5.01%**	-5.19%	-5.22%	9/27	21
pick, crash-guarded: greedy (loss limit, inventory flatten)	-1.64%	+1.13%	**-2.32%**	-2.75%	-2.51%	13/27	18

ENAUSDT: a rebate artefact

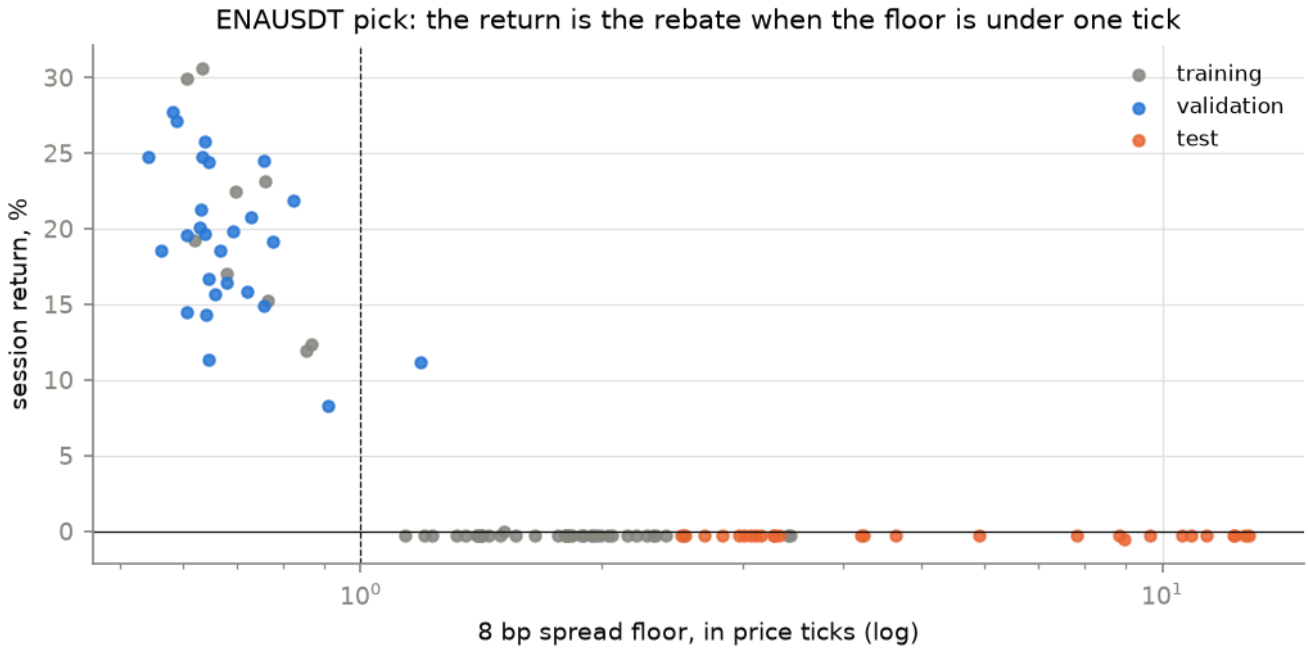
Both rules picked the same configuration (GLFT 8 bp with a stop, momentum, per-side markout and a 25 bp loss limit that flattens).

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, C8's rule: greedy (stop, momentum, per-side markout, loss limit)	+369.67%	+11024.06%	**-6.96%**	-6.73%	-10.02%	0/27	415

entry	training	validation	test	test without its best session	test without the rebate	test sessions up	fills/session (test)
pick, crash-guarded: greedy (stop, momentum, per-side markout, loss limit)	+369.67%	+11024.06%	**-6.96%**	-6.73%	-10.02%	0/27	415

Its return depends on one thing: whether the 8 bp spread floor is under one price tick. Under one tick it quotes at the touch, fills tens of thousands of times a session with almost no drawdown and is paid the 0.5 bp rebate on about 5,000× capital turned over. At one tick or more it loses 25 bp, flattens and stays paused for most of the session.

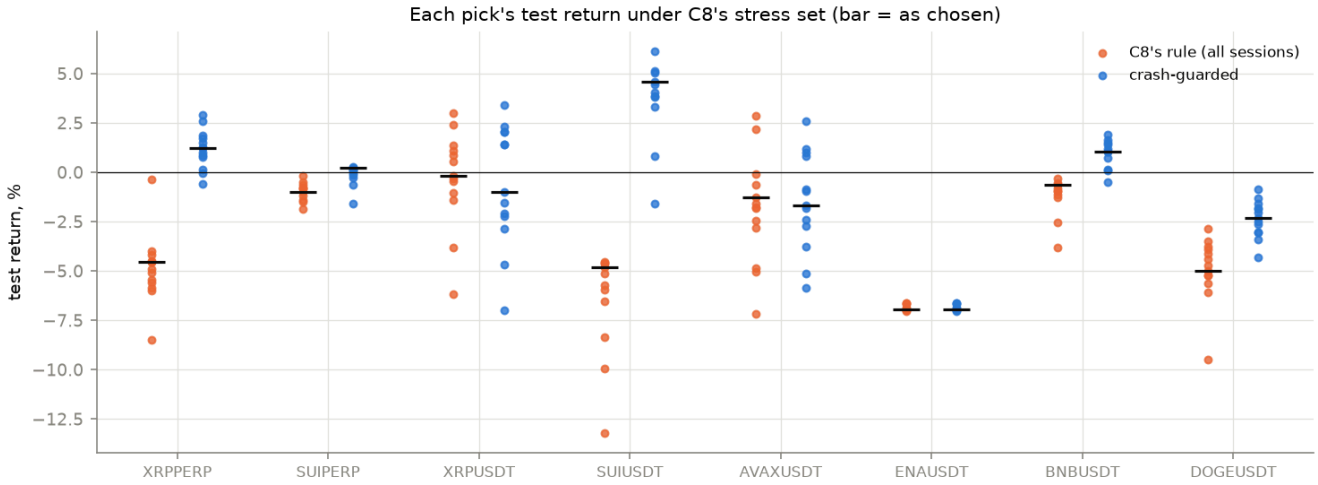
block	sessions	floor under 1 tick	median return, floor < 1 tick	median return, floor ≥ 1 tick	median fills, floor < 1 tick	median fills, floor ≥ 1 tick
training	48	9	+19.23%	-0.25%	78,825	792
validation	27	26	+19.61%	+11.14%	76,183	44,852
test	27	0	–	-0.25%	–	415



The floor was under one tick on 9 training sessions and 26 validation sessions (late January to June 2026), and on no test session. The fee stresses on the test barely move the result (−6.62% to −6.66%) because the quoter has stopped trading. Neither the validation nor the test number says anything about market making on ENAUSDT. What it does say: a backtest with a maker rebate and a queue model that fills a quote at the touch will pay for sub-tick spreads, and a screen should check the floor in ticks.

6. Stress tests, contract by contract

C8's stress set on the test block: maker fee 0 / +1 / +2 bp, latency ×2 and ×4, the risk-averse and log queue models, double size, and the floor and κ neighbourhood (κ rows are absent where the quoter has κ = 0). Each dot is one stress; the bar is the pick as chosen. The figure shows C10's five contracts; the tables below include BNBUSDT and DOGEUSDT.



Crash-guarded picks:

stress (test), crash-guarded	XRPPERP	SUIPERP	XRPUSDT	SUIUSDT	AVAXUSDT	BNBUSDT	DOGEUSDT
as chosen	+1.22%	+0.22%	-1.01%	+4.60%	-1.69%	+1.06%	-2.32%
fee 0	+0.86%	+0.05%	-2.24%	+3.33%	-2.39%	+0.74%	-2.51%
fee +1 bp	+0.14%	-0.29%	-4.66%	+0.84%	-3.76%	+0.11%	-3.04%
fee +2 bp	-0.57%	-0.63%	-7.01%	-1.60%	-5.12%	-0.51%	-3.39%
latency ×2	+1.51%	+0.12%	+3.38%	+4.06%	-0.86%	+1.17%	-0.88%
latency ×4	+2.89%	-1.57%	+2.02%	+3.82%	+2.57%	+1.39%	-3.04%
queue risk-averse	+1.86%	+0.26%	+1.43%	+4.47%	-1.82%	+1.06%	-1.31%
queue log	+0.79%	+0.23%	+1.42%	+3.86%	-2.71%	+1.65%	-1.60%
size ×2	+0.98%	-0.03%	-2.11%	+5.15%	-0.97%	+1.89%	-4.33%
floor ×0.8	+1.73%	-0.14%	-2.85%	+6.14%	+0.83%	+1.49%	-2.65%
floor ×1.25	+1.25%	+0.14%	+2.02%	+5.04%	+0.99%	+0.08%	-1.81%
κ ×0.5	+2.59%	–	-1.56%	–	-5.85%	–	-2.04%
κ ×2	-0.05%	–	+2.30%	–	+1.20%	–	-1.85%

C8's rule picks:

stress (test), C8's rule	XRPPERP	SUIPERP	XRPUSDT	SUIUSDT	AVAXUSDT	BNBUSDT	DOGEUSDT
as chosen	-4.54%	-1.01%	-0.18%	-4.81%	-1.26%	-0.63%	-5.01%
fee 0	-4.88%	-1.18%	-1.41%	-6.56%	-2.47%	-1.27%	-5.22%
fee +1 bp	-5.43%	-1.52%	-3.81%	-9.95%	-4.85%	-2.54%	-5.64%
fee +2 bp	-6.01%	-1.86%	-6.17%	-13.23%	-7.17%	-3.80%	-6.06%
latency ×2	-5.57%	-0.20%	+1.35%	-4.64%	+2.88%	-0.63%	-2.84%
latency ×4	-5.85%	-0.51%	+0.55%	-4.52%	-2.82%	-0.52%	-4.12%
queue risk-averse	-3.99%	-0.80%	-0.45%	-5.96%	-1.80%	-0.80%	-4.73%
queue log	-4.16%	-0.82%	-0.34%	-5.71%	-0.65%	-0.31%	-3.51%
size ×2	-8.49%	-1.39%	+2.99%	-8.34%	-1.77%	-0.95%	-9.50%
floor ×0.8	-4.56%	-0.77%	+1.08%	-4.59%	-0.07%	-0.94%	-3.75%
floor ×1.25	-0.36%	-0.65%	+2.43%	-5.14%	-1.61%	-1.12%	-5.24%
κ ×0.5	-5.08%	–	-1.05%	–	-5.04%	–	-3.91%
κ ×2	-4.47%	–	+0.86%	–	+2.16%	–	-4.40%

- **SUIUSDT, crash-guarded**: positive in 9 of 10 stresses, failing only at +2 bp. The only pick whose result is not within noise of zero.
- **XRPPERP, crash-guarded**: positive in 10 of 12; fails at +2 bp and is flat at κ ×2. About as good as C8's choice unchanged.
- **BNBUSDT, crash-guarded**: positive in 9 of 10; fails at +2 bp (−0.51%), +0.11% at +1 bp.
- **DOGEUSDT**: both picks lose under every stress.
- **XRPUSDT and AVAXUSDT**: the stresses scatter on both sides of zero (AVAXUSDT C8's rule: latency ×2 +2.88%, latency ×4 −2.82%), which is what no edge looks like.
- **Every C8-rule pick except XRPUSDT's loses under nearly every stress.**
- ENAUSDT is left out of these tables (§5).

7. So what?

- **C8's selection rule is the problem on Bybit, not only C8's choices.** Six months of training with a crash in it turns C8's rule into a search for the configuration that held the most inventory on the right side of one rebound. Any rule used to choose on a block with a tail event in it needs a guard on that event; C10's crash guard is enough to stop it.
- **With the guard, the procedure beats running C8's choices unchanged, but narrowly**: +3.33 against +0.64 points summed over C10's five contracts, and all of the margin is SUIUSDT. On the two added contracts with a usable result it makes +1.06% (BNBUSDT) and −2.32% (DOGEUSDT), against −0.63% and −5.01% under C8's rule.

- **The one robust find is a slow grid quoter with a volatility breaker on SUIUSDT.** It keeps its return without the rebate and at +1 bp, under latency and queue stresses, and without its best session. It is the first configuration chosen on Bybit that holds up on a locked test and the stress set (C12's SUIUSDT re-choice won one held-out week, unstressed).
- **BNBUSDT's guarded pick is a smaller, similar find**: a quoter with a breaker, +1.06% on the test, robust to latency, queue and size, positive at +1 bp (+0.11%).
- **A backtest rebate can manufacture a winner.** ENAUSDT's pick earned thousands of percent by quoting under one tick at the rebate; the screen should reject any setting whose spread floor is below one tick.
- **Fees are the binding constraint.** Nothing survives Bybit's standard +2 bp maker fee. Whether any of this is tradeable depends on the fee tier actually available.

8. How it was run, and what went wrong

- Each stage's (configuration, session) backtests ran on Kaggle (five notebooks at a time, 3 workers each, 2 when XRPUSDT sessions are in the batch — more ran out of memory) and were merged into the same per-session cache as C8 and C10. `contest.py units --phase <stage>` dumps the pending work so a fresh notebook needs nothing else.
- The crash guard was added after stage 2 on XRPPERP showed the crash session deciding C8's ranking; both rules then ran on every contract, and both picks went to the test, so the test was not used to choose between the rules.
- The guard does not apply to the screen leader (§4). Fixing it would change the AVAXUSDT, ADAUSDT and LTCUSDT crash-guarded picks; it was left as is so the test stays single-use.
- The fee columns of §3 are recomputed from turnover, which assumes the quoter trades the same at any fee; §5 and §6 use the backtest's own fee stresses. On C10's five the two agree to within 0.02 points; on ENAUSDT they do not (−15.9% recomputed against −6.7% backtested at +1 bp), because the quoter stops trading.
- One pooled batch of Kaggle work was lost to a phase label `contest.py` rejects; it was rerun unchanged.